

MTH-100

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Question No 01 :

$$\left| \frac{3-5i}{5+7i} \right|$$

Solution :

$$\begin{aligned} &= \frac{3-5i}{5+7i} \times \frac{5-7i}{5-7i} \\ &= \frac{(3-5i)(5-7i)}{(5-7i)(5+7i)} \\ &= \frac{15-21i-25i+35i^2}{(5)^2-(7i)^2} \end{aligned}$$

$$\begin{aligned} &= \frac{15-46i+35(-1)}{25-49i^2} \\ &= \frac{-20-46i}{25-49(-1)} \end{aligned}$$

$$i^2 = -1$$

$$= \frac{-20-46i}{25+49}$$

$$= \frac{-20-46i}{74}$$

$$= \frac{-20}{74} - \frac{46}{74}i$$

$$\left| \frac{3-5i}{5+7i} \right| = \sqrt{\left(\frac{-20}{74}\right)^2 + \left(\frac{46}{74}\right)^2}$$

$$= \sqrt{\frac{400}{5476} + \frac{2116}{5476}}$$

$$= \sqrt{\frac{2516}{5476}}$$

Question No 02

Determine whether the function $f(x) = 2/x$ is even or odd. Also, find its domain and range.

Answer:

To determine whether the function $f(x) = \frac{2}{x}$ is even or odd, we can use the definitions:

- A function is even if $f(-x) = f(x)$ for all x
- A function is odd if $f(-x) = -f(x)$ for all x

Step 1: Check if the function is even or odd

1. Calculate $f(-x)$

$$f(-x) = \frac{2}{-x} = -\frac{2}{x}$$

2. Compare $f(-x)$ with $-f(x)$

$$-f(x) = -\left(\frac{2}{x}\right) = -\frac{2}{x}$$

Since $f(-x) = -f(x)$ the function is odd.

Step 2: Determine the domain

The function $f(x) = \frac{2}{x}$ is defined for all real numbers except where the denominator is zero. Thus, the domain is:

Domain: $(-\infty, 0) \cup (0, \infty)$

Step 3: Determine the range

The function $f(x) = 2/x$ can take on all real values except for zero. As x approaches 0 from the left, $F(x)$ approaches ∞ , and as x approaches 0 from the right, $f(x)$ approaches $+\infty$

Therefore, the range is:

Range : $(-\infty, 0) \cup (0, \infty)$

Summary

- The function $f(x) = 2/x$ is odd.
- Domain: $(-\infty, 0) \cup (0, \infty)$
- Range: $(-\infty, 0) \cup (0, \infty)$

Question No 03

Let $f(x) = 5x + 3$ and $g(x) = 3x - 5$ be the two functions then find the composite

Functions $(f \circ g)(x)$ and $(g \circ f)(x)$.

Answer:

First = $(f \circ g)(x)$

$F(g(x)) = f(3x-5)$

$F(3x-5) = 5(3x-5) + 3$

$$= 15x - 25 + 3$$

$$= 15x - 22$$

—→ Now: $(g \circ f)(x)$

$$g(f(x)) = g(5x + 3)$$

$$= (3(5x + 3) - 5)$$

$$= 15x + 9 - 5$$

$$= 15x + 4$$

Question 04

Solve the following system of equations by applying row operation.

$$3x_1 + 3x_2 = 15$$

$$15x_1 + 16x_2 = 23$$

Answer:

To solve the system of equations using row operations, let's first write down the system in matrix form:

The system is:

$$3x_1 + 3x_2 = 15$$

$$15x_1 + 16x_2 = 23$$

In matrix form, this is:

$$\begin{bmatrix} 3 & 3 \\ 15 & 16 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 15 \\ 23 \end{bmatrix}$$

Now, we'll apply row operations to solve for x_1 and x_2

Step 1: Form the augmented matrix:

$$\left(\begin{array}{cc|c} 3 & 3 & 15 \\ 15 & 16 & 23 \end{array} \right)$$

Step 2: Make the leading coefficient of the first row equal to 1

To make the leading coefficient in the first row a 1, divide the entire first row by 3:

$$\left(\begin{array}{cc|c} 1 & 1 & 5 \\ 15 & 16 & 23 \end{array} \right)$$

Step 3: Eliminate the x_1 term in the second row

To make the x_1 coefficient in the second row 0, subtract 15 times the first row from the second row:

$$R_2 = R_2 - 15R_1$$

$$\left(\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & 1 & -52 \end{array} \right)$$

Step 4: Substitute back to find x_1 and x_2

Now we have:

$$x_1 + x_2 = 5$$

$$x_2 = -52$$

From the second equation, we get $x_2 = -52$ substitutes $x_2 = -52$ into the first equation:

$$x_1 + (-52) = 5$$

$$x_1 = 5 + 52 = 57$$

Solution

The solution to the system is:

$$x_1 = 57, x_2 = -52$$

